

Sudeep Bhimannavar

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SUMMARY

Software Engineer with production experience building AI-powered systems and distributed SaaS platforms at scale. Architected AI-driven indoor navigation using computer vision and graph algorithms, scaled RESTful APIs to **50,000+ requests/day at 99.9% uptime**, and shipped multi-tenant platforms serving **10,000+ daily active users**. Expert in Python, Django, React.js, and MySQL, with a track record of cutting query latency by **60%** and enforcing code quality through peer reviews and TDD in Agile environments.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), Java, C, SQL, HTML5, CSS3

Frameworks & Libraries: Django, Django REST Framework, React.js, NetworkX, NumPy, Pandas, OpenCV

Databases & Cloud: MySQL, PostgreSQL, Firebase, AWS (Basics), Railway

Infrastructure & Tools: Git, GitHub, Docker, Postman, CI/CD, JWT, System Design, TDD

AI/ML & Engineering: Claude Vision API, Generative AI, Computer Vision, Dijkstra's Algorithm, Graph Theory, NLP

Accessibility: WCAG 2.1 AA, ARIA Landmark Roles, Semantic HTML

EXPERIENCE

Software Engineer | Vividhity Ventures Pvt. Ltd., Bengaluru

July 2025 – Present

- Architected and shipped **Formalls**, a production multi tenant SaaS platform managing 100+ stores, 20+ restaurants, and 500+ daily bookings across 5+ enterprise clients with strict cross tenant data isolation.
- Scaled backend infrastructure to sustain **50,000+ daily API requests at 99.9% uptime**, redesigned SQL indexing and schema architecture, cutting average response latency by **60%**.
- Delivered **MallNav**, an AI driven indoor navigation system supporting **1,000+ concurrent users** at 95% route accuracy, compressing environment setup from 8 hours to 45 minutes.
- Built a collaborative filtering recommendation engine leveraging user visit and purchase history, driving a **30% increase in engagement** and a **25% increase in average transaction value**.
- Integrated WhatsApp Business API for round the clock customer support, cutting ticket volume by **45%** while sustaining a **95% CSAT score**.
- Enforced **zero leakage** tenant security for **10,000+ users** via a 5 tier JWT based RBAC system across all client boundaries.
- Maintained **85%+ test coverage** through TDD and structured peer reviews within an 8 member Agile team, authored comprehensive API documentation adopted across all product squads.

PROJECTS

MallNav: AI Driven Indoor Navigation Architecture

Feb 2026 – Present

Python, Django, NetworkX, OpenCV

- Built a Claude Vision API + OpenCV pipeline to autonomously parse architectural floor plans, classifying Points of Interest with **90% accuracy** and zero manual intervention (graph based model chosen over raster for dynamic weight updates).
- Designed a multi floor pathfinding engine using **Dijkstra's Algorithm** on dynamic weighted graphs, paired with a custom HTML5 Canvas editor supporting real time edge weight tuning and node snapping.
- Deployed a QR based, installation free navigation experience for **1,000+ concurrent users**, automated POI mapping slashed environment setup time by **90%** — from 8 hours to 45 minutes.
- Authored technical documentation and map maintenance workflows enabling non technical staff to self serve floor plan updates, yielding an **80% reduction** in engineering support requests.

Formalls: Production Multi tenant SaaS Infrastructure

July 2025 – Present

Django, React.js, MySQL, JWT

- Designed a normalized multi tenant MySQL schema with strict data isolation across 100+ retail locations (row level tenancy chosen over schema per tenant), engineered high concurrency booking flows with real time slot conflict detection.
- Built an Events & Promotions module with multichannel notification dispatch and 5 tier JWT based RBAC, sustaining **99.9% uptime** through peak promotional traffic without degradation.
- Overhauled the frontend to achieve full **WCAG 2.1 AA** accessibility compliance and mobile first responsiveness, reducing manual mall operations overhead by **70%**.

EDUCATION

Master of Computer Applications (MCA) — Distance Learning

2025 – 2027

Jain University, Bengaluru, India | Relevant Coursework: Machine Learning, Deep Learning, NLP, System Design, Algorithms & Data Structures, Database Management, Computer Networks

Bachelor of Computer Applications (BCA)

2020 – 2023

Bharatesh College of Computer Applications, Belagavi, Karnataka